Ernest Choi

ernestljchoi@gmail.com | github.com/ErChoi-com | linkedin.com/in/ernest-choi-lj | 647-713-8997

Projects

eebot Mobile Robot System | *C, Assembly, HCS12, PWM*

- Developed embedded firmware in **Assembly** and **C** for a mobile robot built on the **HCS12** microcontroller, handling hardware verification and board-level integration in a Linux/Unix environment.
- Implemented **PWM** motor control with timer-driven scheduling for mobile robot electronics, sustaining 50ms sensor sampling intervals to meet technical performance deadlines.

General-Purpose Processor (ALU) | *VHDL, FPGA, Quartus II, RTL*

- Designed and implemented a modular Arithmetic and Logic Unit (ALU) in **VHDL** on an **Altera FPGA**, integrating registers, a finite state machine, a 4×16 decoder, and a microcode-driven ALU core.
- Verified digital logic designs through **Quartus II** simulation, waveform analysis, and static timing verification, ensuring quality standards were met for hardware.
- Applied **RTL** design practice — timing analysis, modular integration, bitwise operations — to build and test digital logic at the register-transfer level.

Experience

Electrical Team Member, Metropolitan Aerospace Rocket Society (MARS) – Toronto, ON September 2023 – Present

- Designed, fabricated, and integrated **PCB** circuits using **Altium** and **KiCad** for custom STM32-based avionics flight computer with GPS telemetry, sensor integration, and pyrotechnic ignition systems
- Optimized circuit configurations and **power distribution** to enhance power efficiency by 23%, applying analysis to component selection for robust electrical systems.
- Performed board bring-up, debugged electronic circuit boards using **oscilloscopes** and **logic analyzers**, and validated hardware functionality against strict quality standards.
- Developed embedded firmware in **C/C++** and automated testing workflows with **Python** scripting for technical avionics systems.

Engineering Instructor, Obotz Robotics – Markham, Ontario September 2022 – Present

- Taught digital logic concepts and embedded **C** programming, fostering problem-solving skills and raising student robotics problem-completion scores by 35%.
- Coached students in designing and building custom **Arduino** hardware for robotics, leading to 75th percentile placements in technical competitions.
- Communicated students' technical progress to parents in clear, accessible language.

Technical Development Intern, City of Markham – Markham, Ontario July 2022 – August 2024

- Maintained clear documentation for website workflows, contributing to continuous improvement and reducing website downtime by 41%.
- Tracked project milestones and reported progress to management using Microsoft Office, which reduced coordination delays by 14% and supported production workflows.

Skills

Languages: JavaScript/Node.js, Java, Python, C++, C, SQL, HTML, CSS
Frameworks/Libraries: OpenGL, WebRTC, JavaFX, React, TypeScript, Flask, FastAPI, TensorFlow
Tools: Microsoft Suite, Git, PostgreSQL, Ubuntu Linux, Docker, Kafka, Tableau, Power BI
Platforms: GitHub, GitLab, Raspberry Pi, Arduino
Relevant Courses: Object Oriented Eng Analysis and Design, Electronic Circuits I, Digital Systems, Microprocessor Systems

Education

Toronto Metropolitan University – Bachelor of Engineering in Computer Engineering 2023–2027